

Nursery Stock Production

Science

May, 2026



Nursery Stock Production (A08541)

Short Title: Nursery Stock Production
Faculty Science and Computing
Department Science
Cluster Horticulture
Author PFITTERS
Credits 10

Level: Intermediate

Description of Module / Aims

This module is designed to equip students with knowledge of nursery stock production, management and planning. Students will learn nursery stock specific skills of planting, potting, crop protection and will perform cultural techniques for bedding plant, trees and shrub production. Plant health and exporting regulations are also covered.

Programmes

HORT-0023	BSc in Horticulture (WD_SHORT_D)	2 / 3 / E
HORT-0023	BSc in Horticulture (WD_SHORT_D)	3 / 5 / E
HORT-0023	BSc in Horticulture (WD_SHORT_DX)	2 / 3 / E
HORT-0023	BSc in Horticulture (WD_SHORT_DX)	3 / 5 / E

Indicative Content

- Bedding plants: Nutritional programmes, Production of a range of bedding plants, Plan a crop protection programme
- Liner production: Sources of material, Propagation systems, Perform cultural techniques, Nutritional programmes
- Containerised trees, shrubs and herbaceous plants: Compost specification for various production programmes, Demonstrate potting skills, Growing-on systems, Plan a crop protection programme, Perform cultural techniques
- Open ground trees and shrubs: Soil preparation operations before planting, Budding and grafting for field tree production, Planting techniques, Plan a crop protection programme, Perform cultural techniques, Harvesting techniques
- Nursery stock management: Plan the physical layout of a nursery, Schedule production programmes of the above, Nursery irrigation systems, Monitor crop performance
- Quality assurance schemes: Quality standard specification for bedding, liners, trees, shrubs and herbaceous plants
- Sustainable plant production systems and use of contemporary new developments.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Demonstrate in a safe working environment, a range of cultural tasks associated with nursery stock production including, budding, grafting, potting and planting.
2. Appraise the production of nursery stock crops such as bedding plants, liners, containerised plants and open ground stock.
3. Propose a crop protection programme appropriate to the nursery stock crop grown e.g. bedding plants, liners, containerised plants and open ground production systems.
4. Determine appropriate nutritional programs and compost specifications for a nursery stock crop.
5. Appraise a range of irrigation methods.
6. Investigate the effects of various propagation environments and materials for the production of liners.
7. Relate the requirements of the nursery stock quality assurance schemes and standard specifications to management strategies.

Learning and Teaching Methods

- Lectures
- Projects
- Practicals
- Site visits

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	2,3,5,7
Continuous Assessment	50%	
<i>Practical</i>	30%	1
<i>Assignment</i>	20%	4,6

Assessment Criteria

- <40%:** Very limited knowledge of plant production methods and cannot demonstrate understanding of plant production. Unable to complete practical skills or exercises.
- 40%-49%:** Demonstrates a limited knowledge of plant production systems and processes. Limited technical ability but demonstrates ability to learn and understand new skills.
- 50%-59%:** Demonstrates satisfactory general knowledge of the main areas of nursery stock production. Competently carries out practical tasks and skills. Reports and projects adequately structured and referenced.
- 60%-69%:** Demonstrates sound knowledge of nursery stock production principles and systems. Demonstrates ability to interpret and analyse technical information logically. Projects, reports and exams coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.
- 70%-100%:** Demonstrates authoritative handling of complex material and a highly developed and extensive understanding of the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical/ professional tasks.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	48	
Practical	36	
Field Trips	12	
Independent Learning	174	

Supplementary Material

- Davidson, H., R. Mecklenburg and C. Peterson. _Nursery management: administration and culture_. 4th Edition. UK: Prentice hall, 2000.
- Lamb, K., J. Kelly and P. Bowbrick. _Nursery stock manual_. England: Grower books, 1995.
- Perry, L.P. _Herbaceous perennials__production: A guide from propagation to marketing_. Vermont: NARES series, 1998.
- Whitcomb, C. _Plant production in containers_. USA: Lacebark, 2003.